

From Windows to Watts — Evolution of Proxies



Window Tax, 1696

Physical · One-dimensional · Binary



Postal Code, 1850

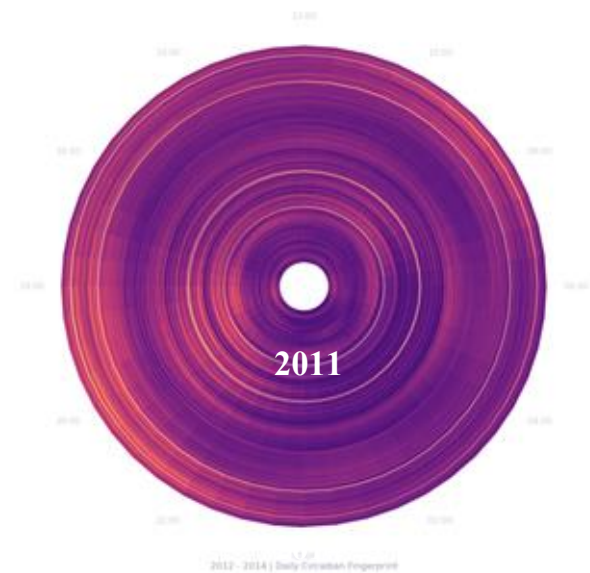
Geographic · Categorical · Static



‘Acorn’

Socioeconomic · Multi-dimensional · Survey-based

One household, three years of electricity, coiled into a single form — color intensity maps consumption.



‘Smart Meters in London’

Behavioral · Temporal · Continuous

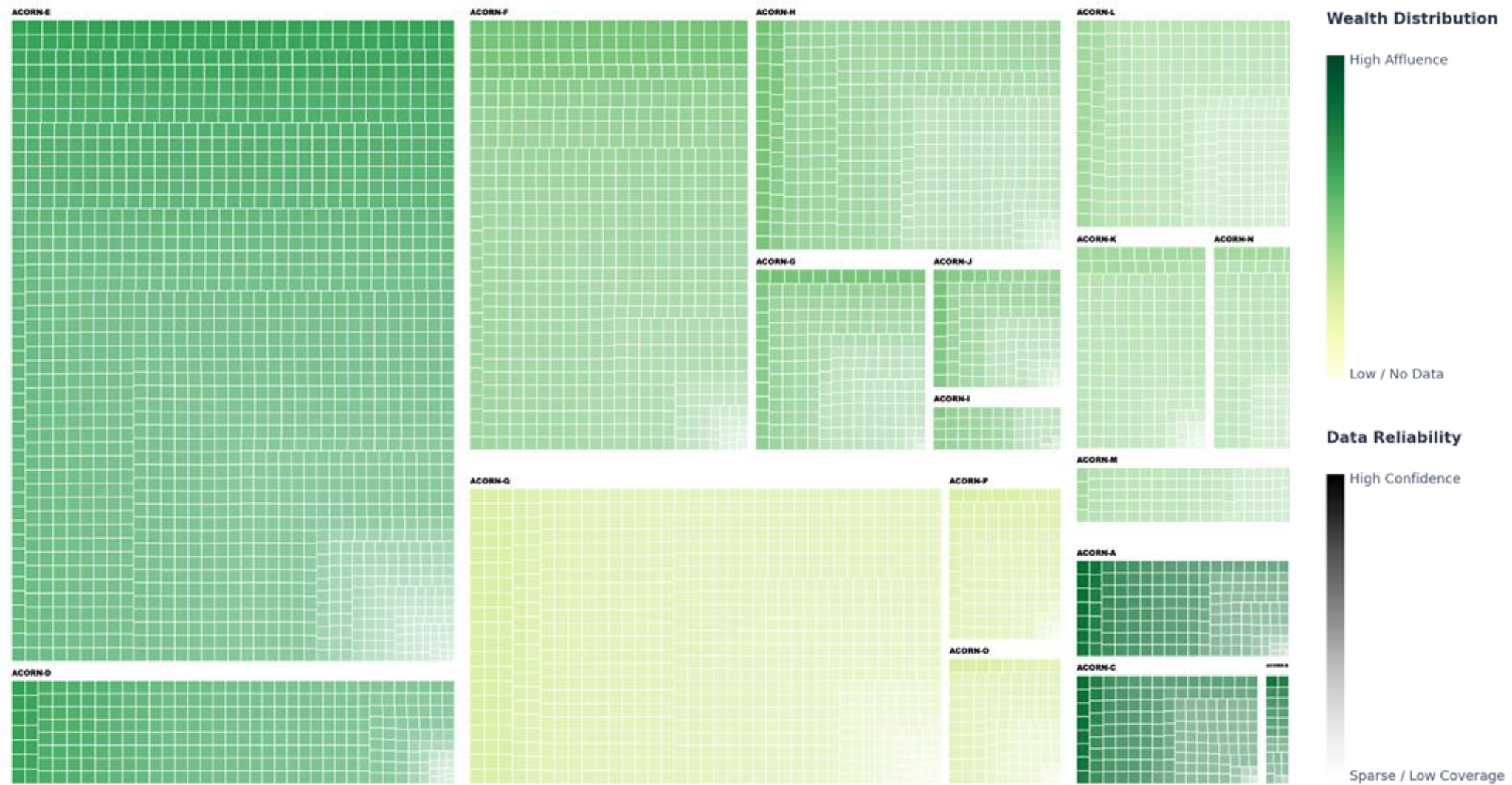
Static



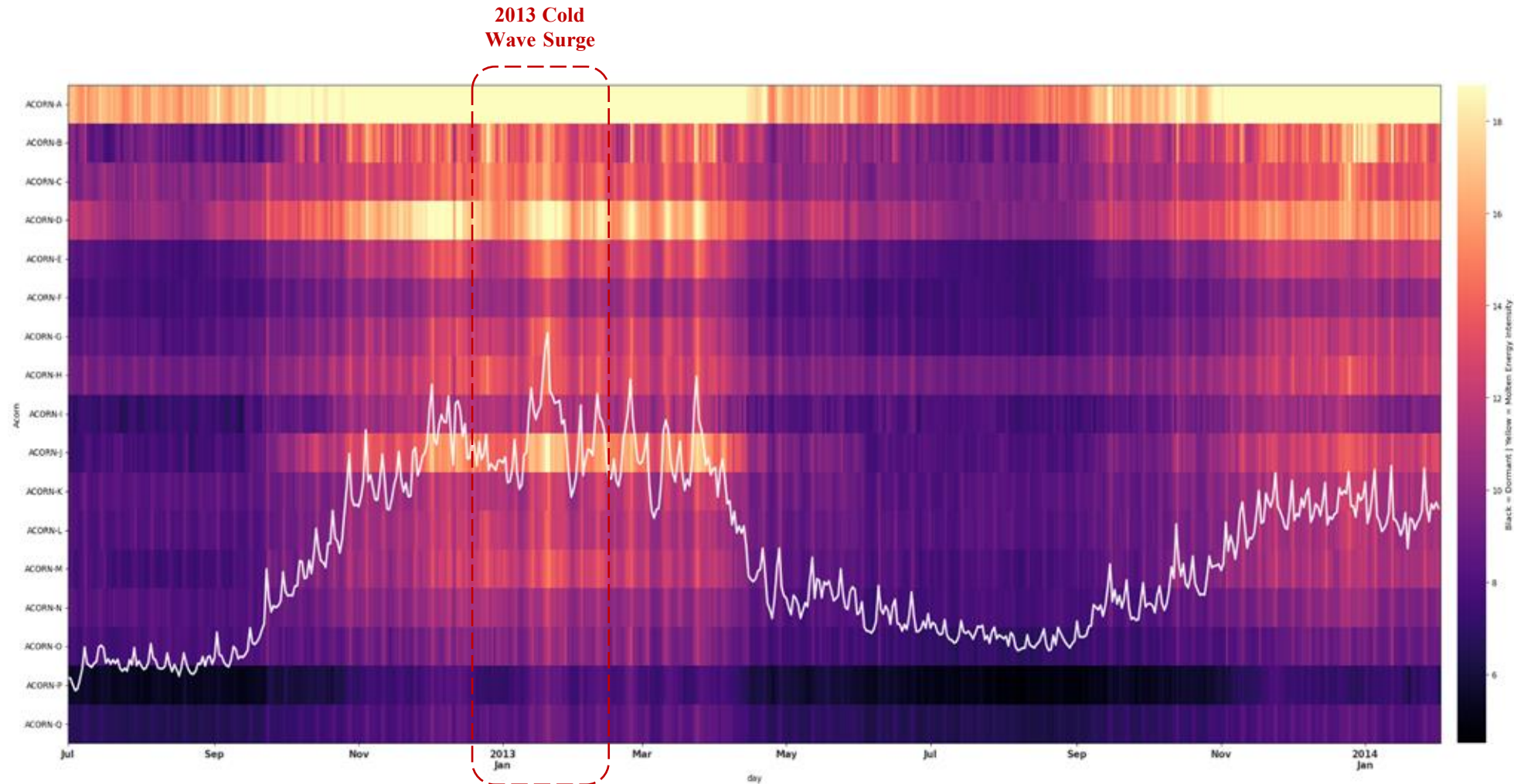
Dynamic

Data Landscape

5,556 London households, each square one home — color maps wealth, opacity maps data confidence.

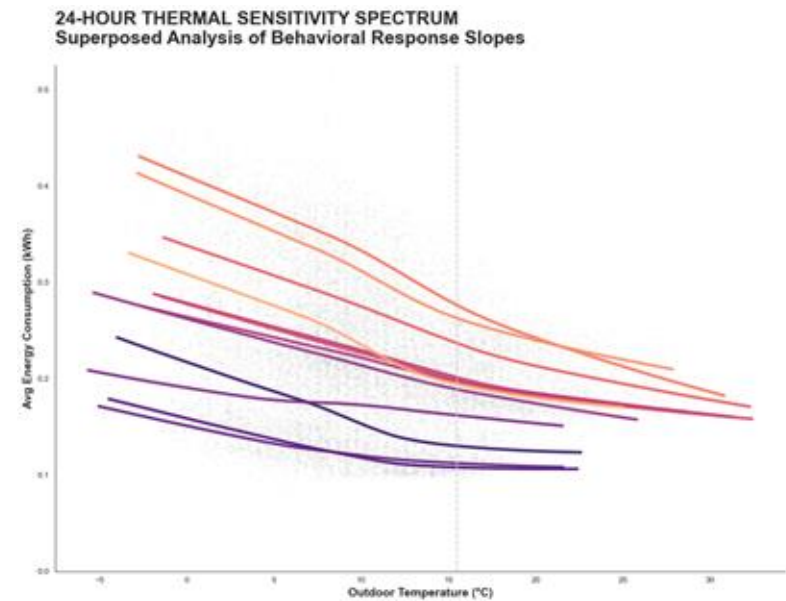
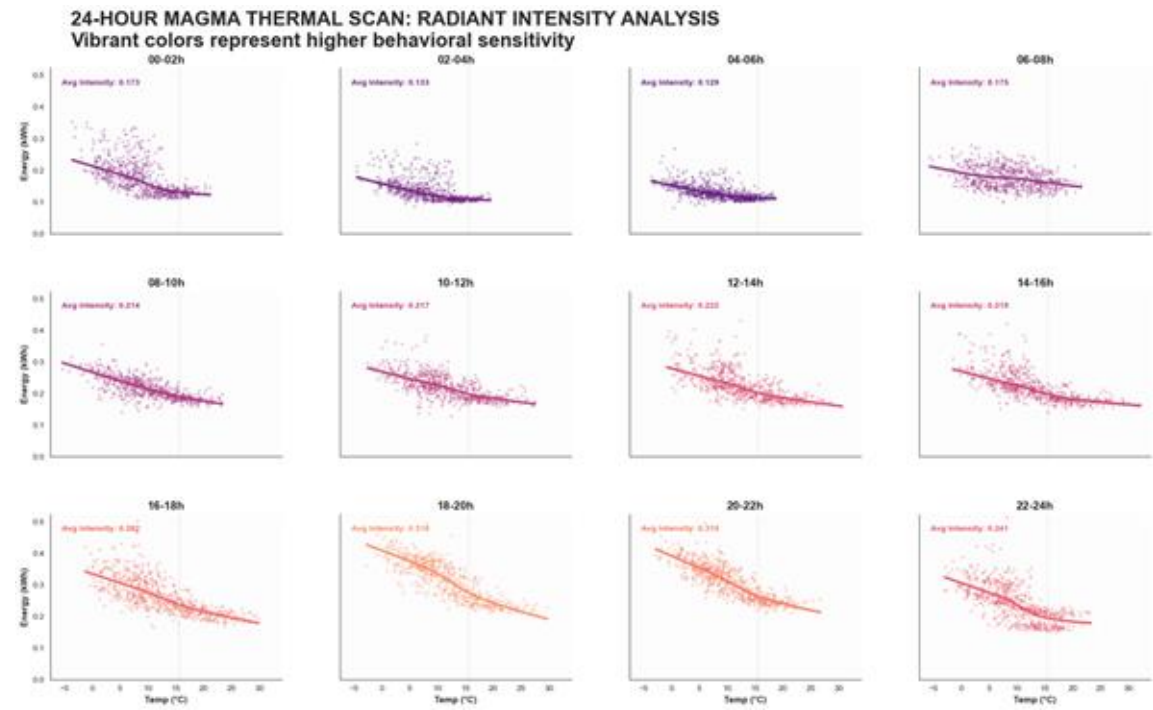


18 Months of London Energy



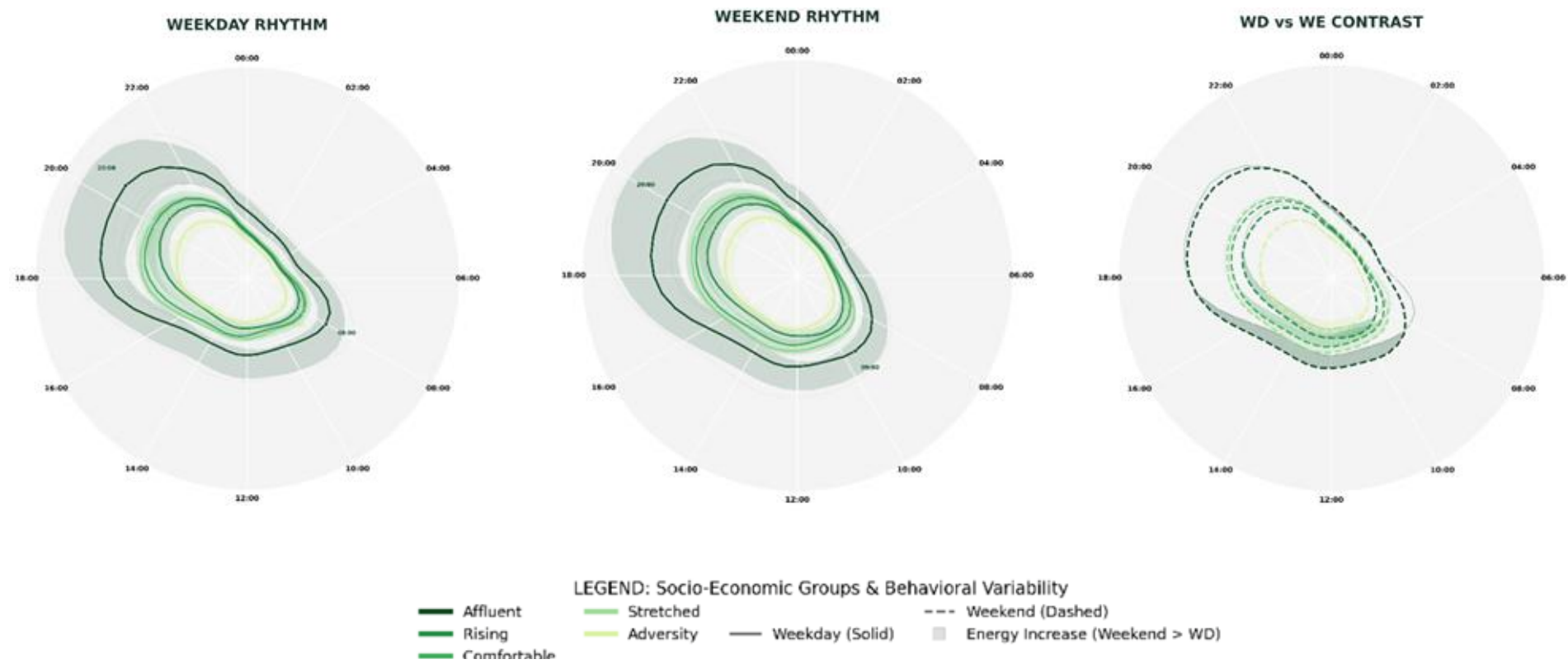
- 1. Extreme Events:** Intense yellow cluster captures early 2013 historic cold wave.
- 2. Weekly Rhythm:** Vertical striations every 7 days mark distinct weekend peaks.
- 3. Stratification:** Affluent groups float above the mean; Adversity groups are compressed below.

24 Hours of Thermal Response



Evening hours (18–22h) show the steepest thermal response — cold weather hits hardest when people are home.
Night hours stay flat; daytime sits in between.

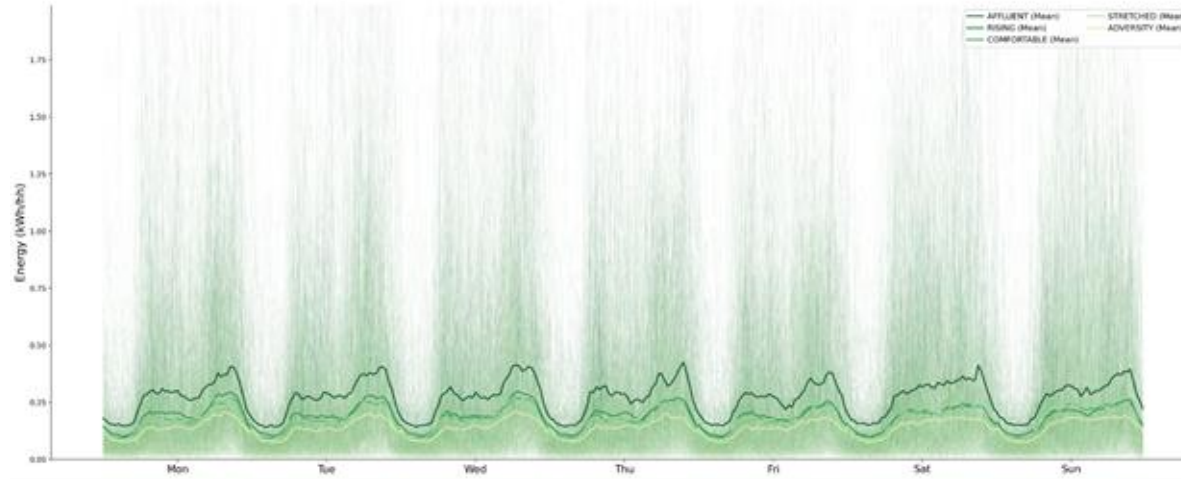
Weekday vs. Weekend



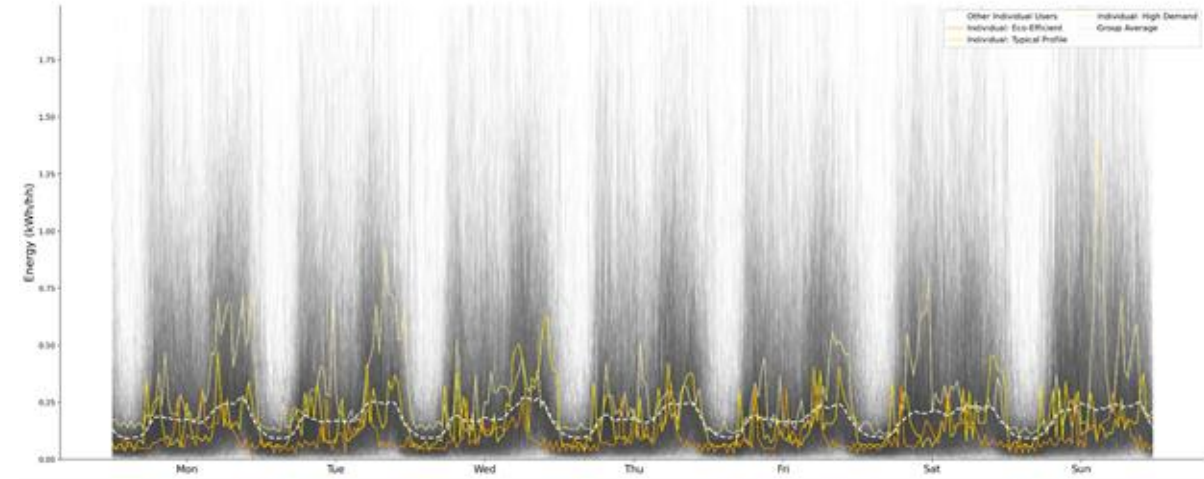
Weekdays show twin peaks at 07:00 and 19:00; weekends the morning peak shifts past 09:00 — the grey area on the right measures the extra energy from staying home.

Labels Sort Groups, Not Homes

SOCIO-ECONOMIC ENERGY FINGERPRINTS



INDIVIDUAL BEHAVIOR SPECTRUM: COMFORTABLE



Unpacking
“Comfortable”

Social labels separate group averages cleanly — but zoom into any single group and the average line becomes a fiction. Real behavioral diversity lives at the individual level, which is why we need **clustering**.

Feature Construction From Meter to Meaning

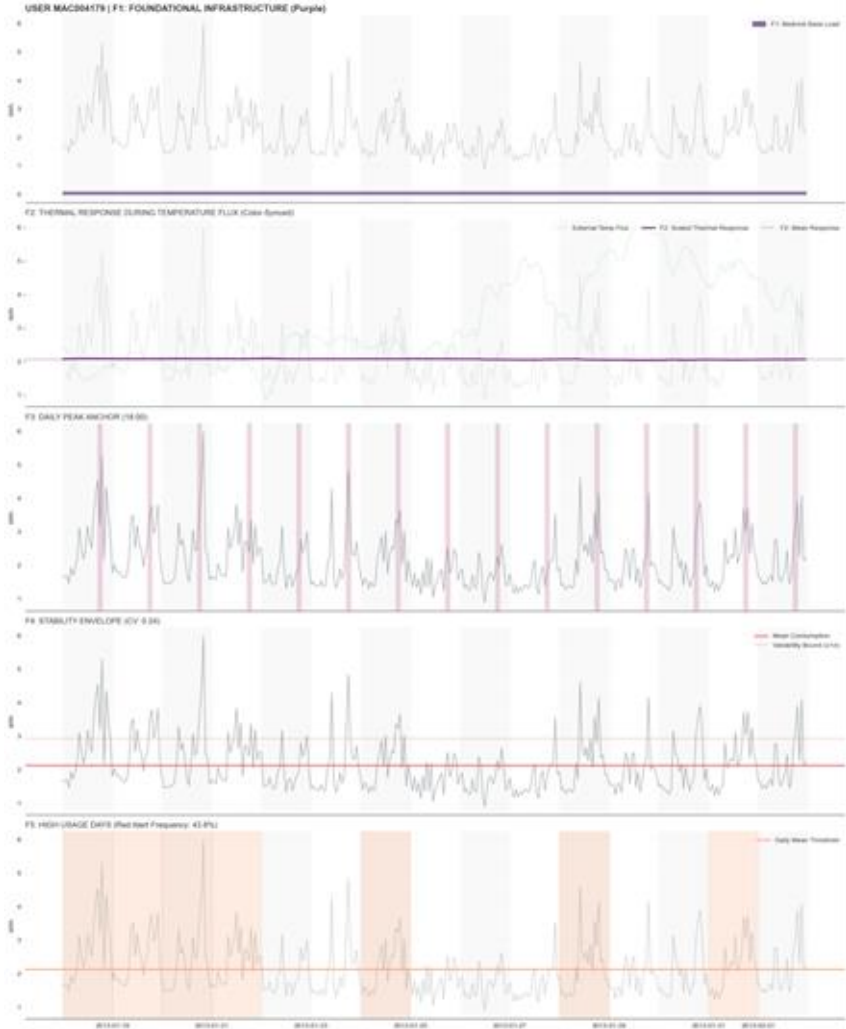
F1

F2

F3

F4

F5



1.Feature Extraction

F1: Baseload

$E_{base} = \min(E_{daily})$

The Floor:
Constant standby power.

F2: Thermal

$\beta = \Delta E / \Delta T$

The Response:
Energy rise per 1°C drop.

F3: Anchor

$T_{peak} = \operatorname{argmax}(E_{hour})$

The Rhythm:
Peak consumption hour.

F4: Stability

$CV = \sigma / \mu$

The Pattern:
Fluctuation vs. Average.

F5: Frequency

$\text{Freq} = \text{Count}(\text{High}) / \text{Total}$

The Event:
Probability of extreme usage.

2.Transforming & Scaling

A: Log
B: Yeo-Johnson

+

Standardization

3.Clustering

K Selection :
Elbow Method | Silhouette

4.Evaluation

Internal Metrics :
Silhouette | CH Index | DB Index

Dimensionality Reduction :
t-SNE Visualization

Feature Evaluation

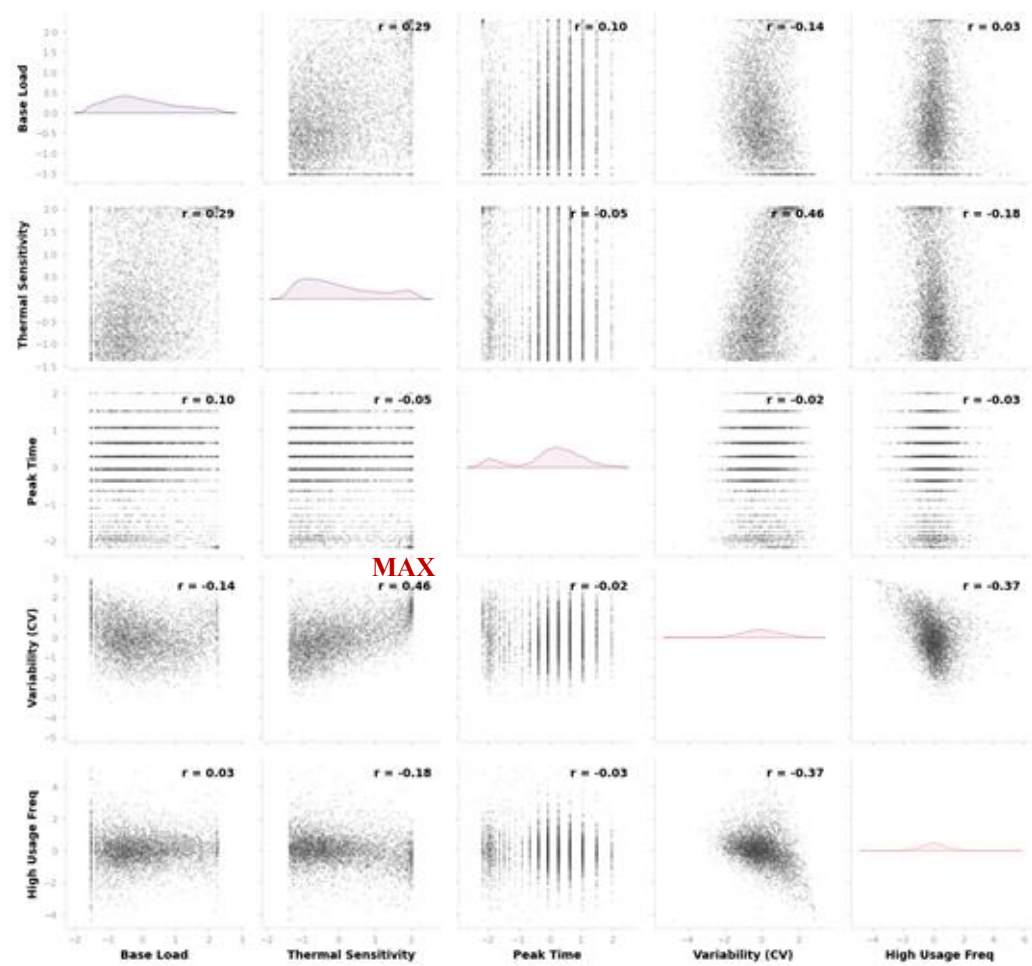
The Skewness Challenge: Raw energy data exhibits heavy tails (Skew > 2.0), which disproportionately biases cluster centroids toward outliers.

Pathway A (Yeo-Johnson): Stabilizes variance and achieves Gaussian-like distribution; ensures more balanced and stable cluster partitions.

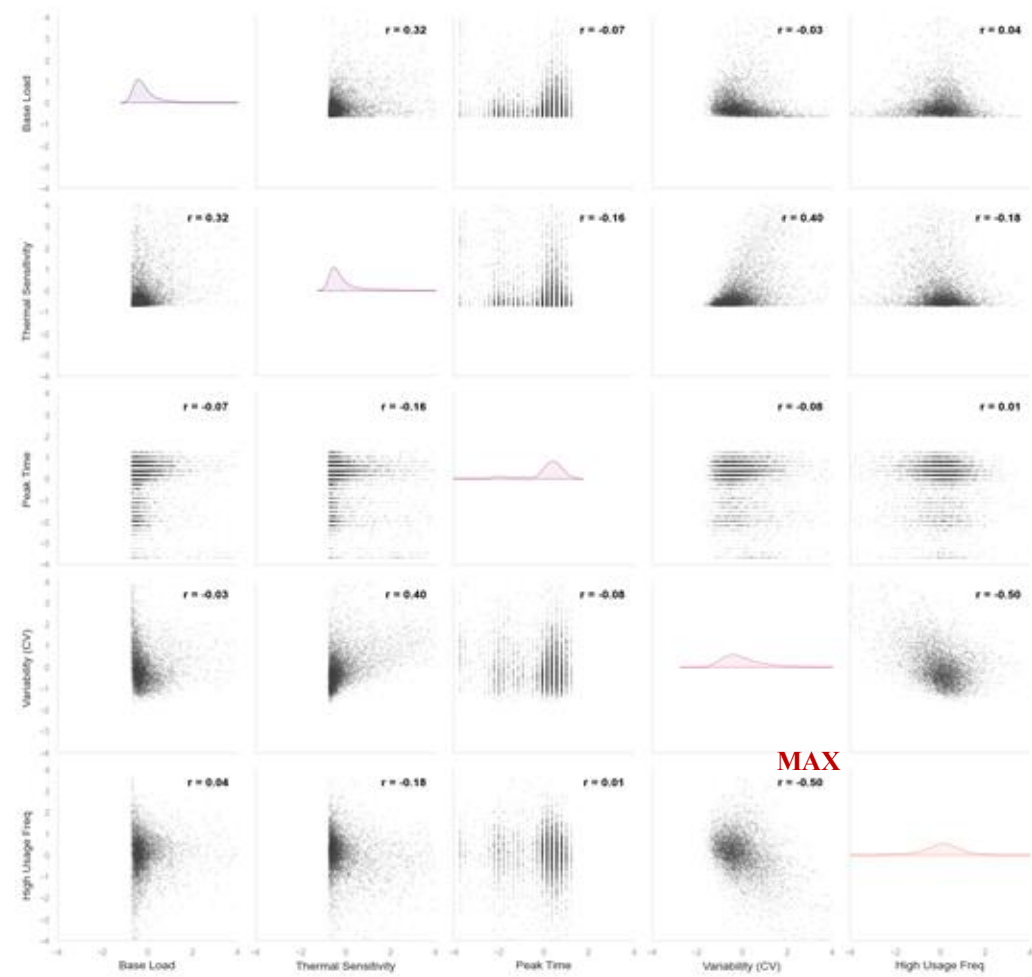
Pathway B (Log Transform): Preserves "Long-tail" information; ideal for **Anomaly Detection** but maintains non-normal distribution.



Pairwise Scatter Matrix with Correlation Audit

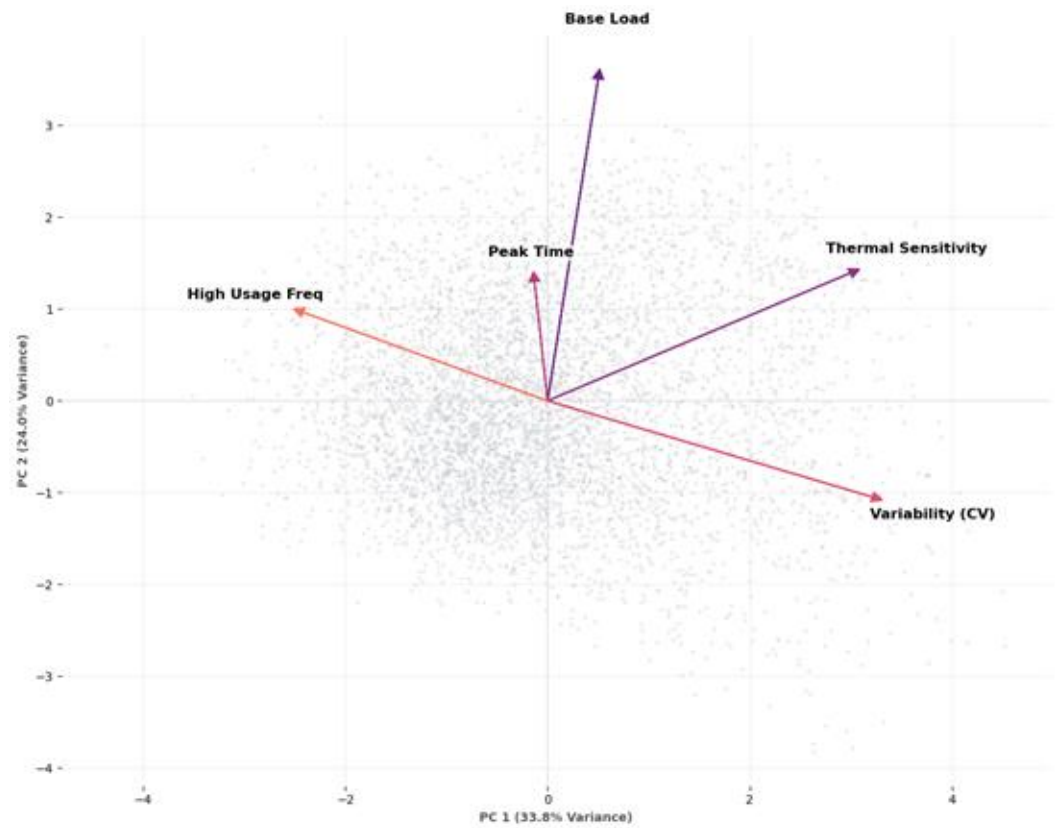


A: Yeo-Johnson Transform + Standardization

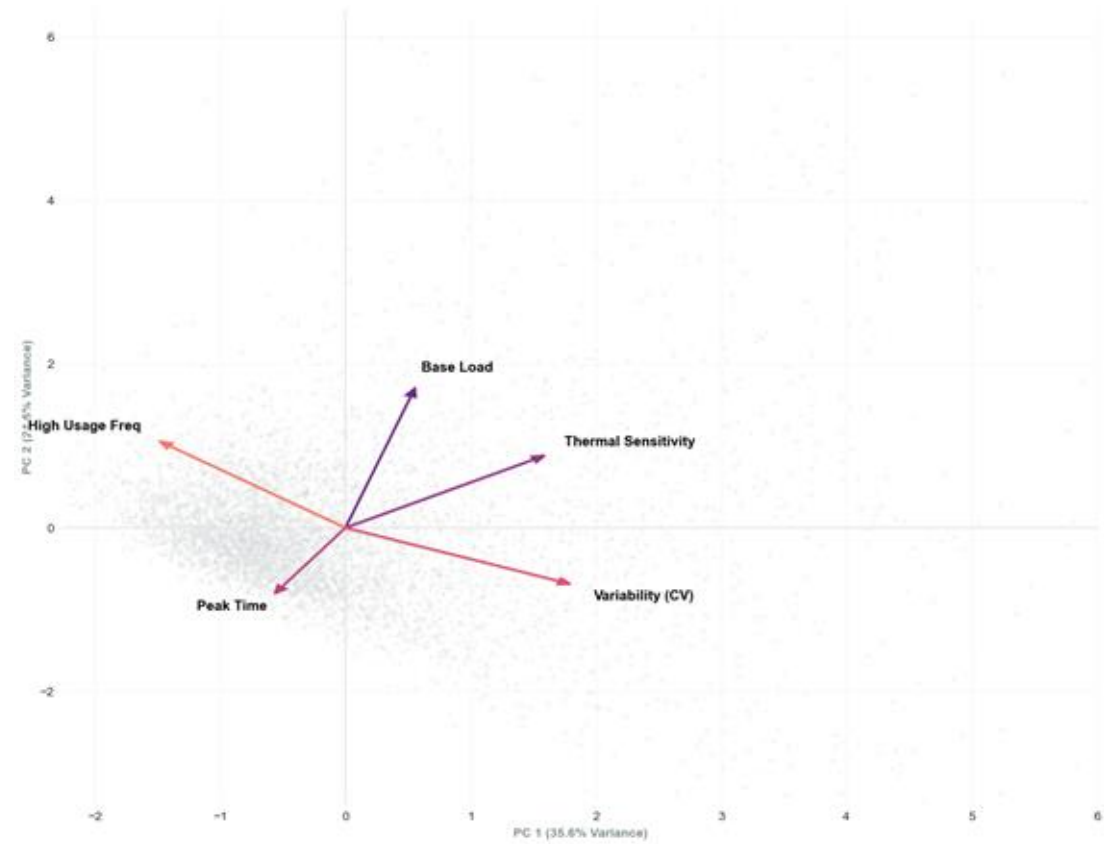


B: Log Transform + Standardization

PCA Structure Audit

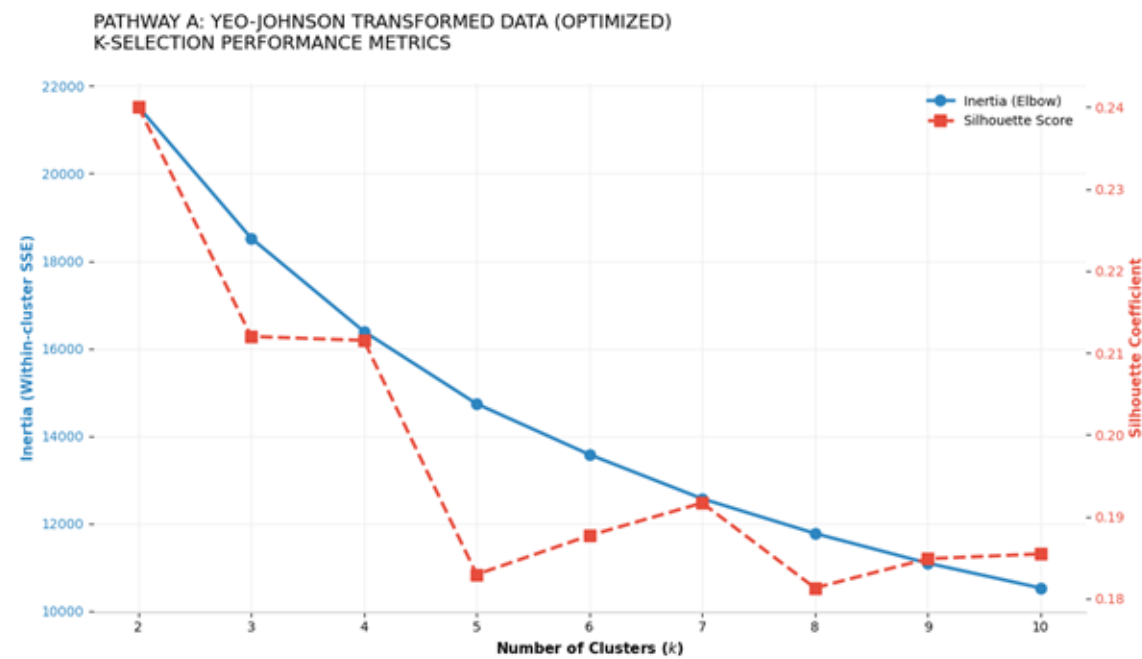


A: YJ Scaling
Data appears uniform and centered



B: Log-Transform
Data shows a dynamic, irregular density

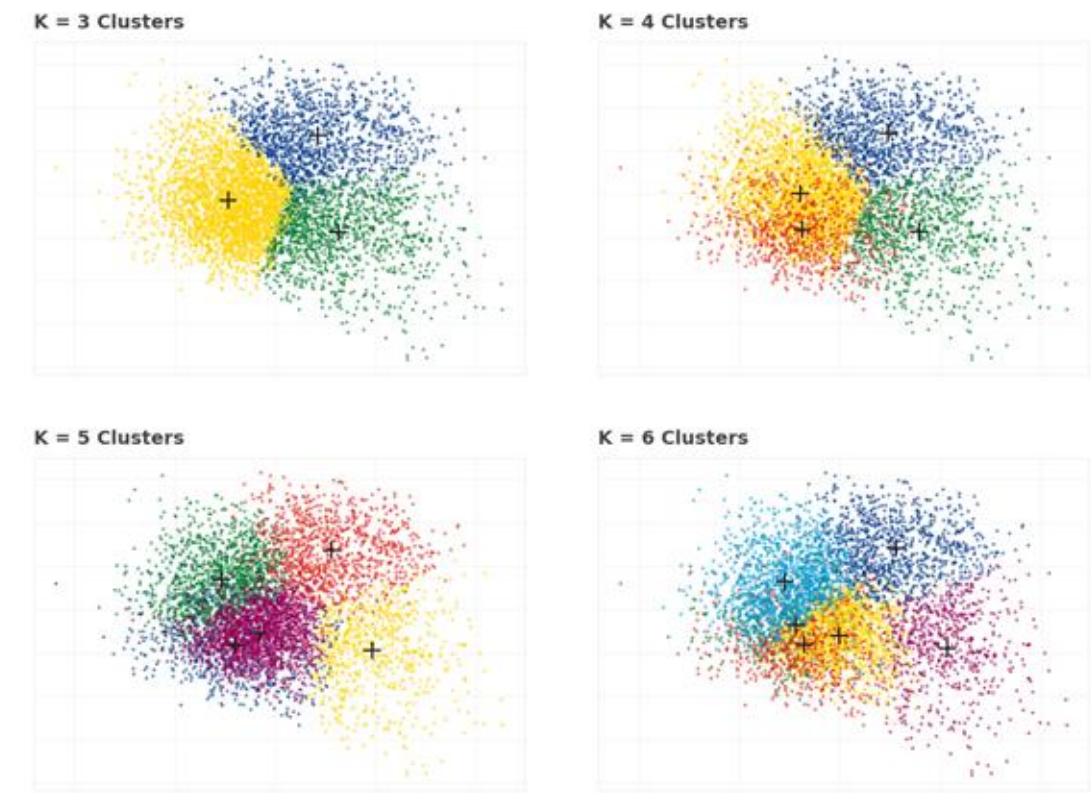
K Selection — Pathway A (Yeo-Johnson)



The Elbow (K=4):

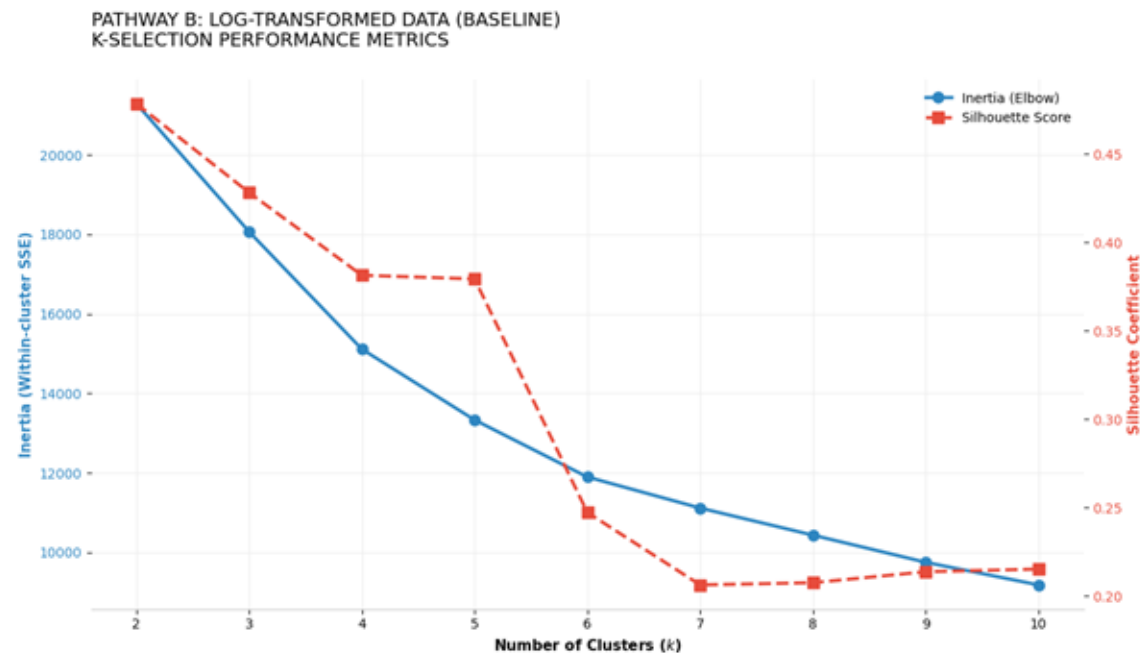
The inertia curve bends gradually; K=4 offers a reasonable balance between cluster count and within-cluster variance. Silhouette score plateaus from K=3 onward, supporting K=4 as a practical choice.

PATHWAY A (YJ): RAW DATA TOPOLOGY EVOLUTION



Yeo-Johnson smooths the distribution — clusters show soft boundaries with visible overlap.

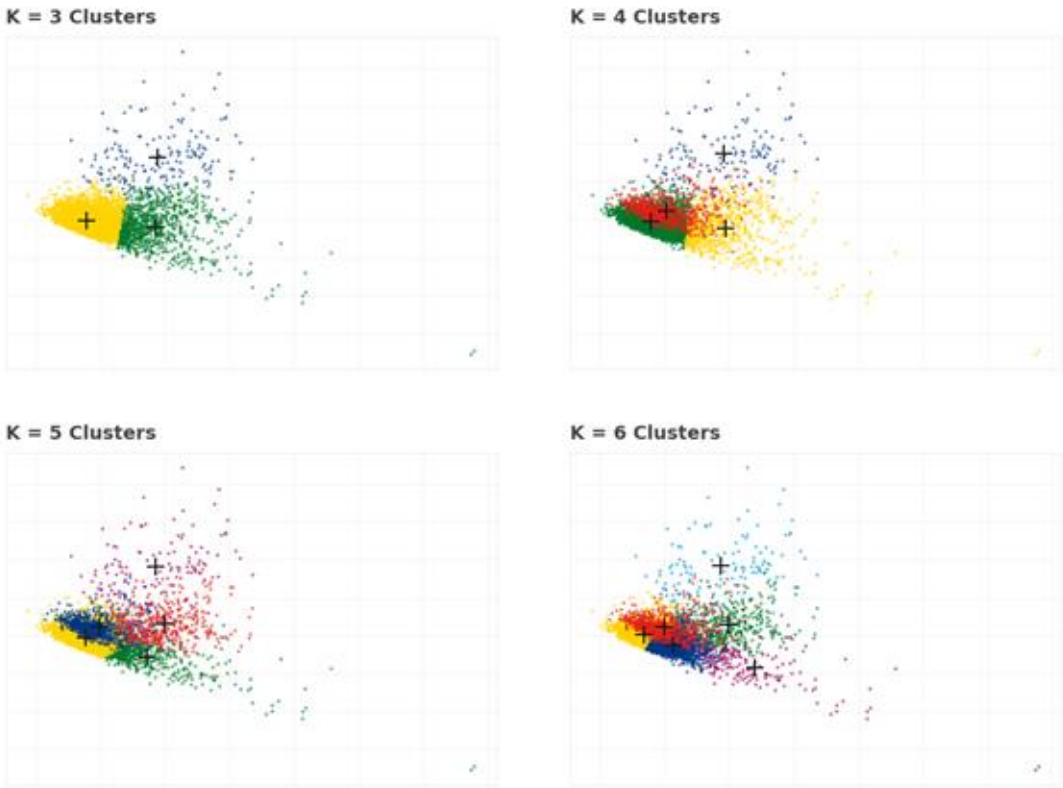
K Selection — Pathway B (Log Transform)



The Elbow (K=5&4):

Mathematically the elbow falls at K=5, but the marginal gain over K=4 is minimal — Silhouette scores near-identical. K=4 already captures the core behavioral archetypes; a fifth cluster subdivides without adding interpretive value. We proceed with K=4 for both pathways.

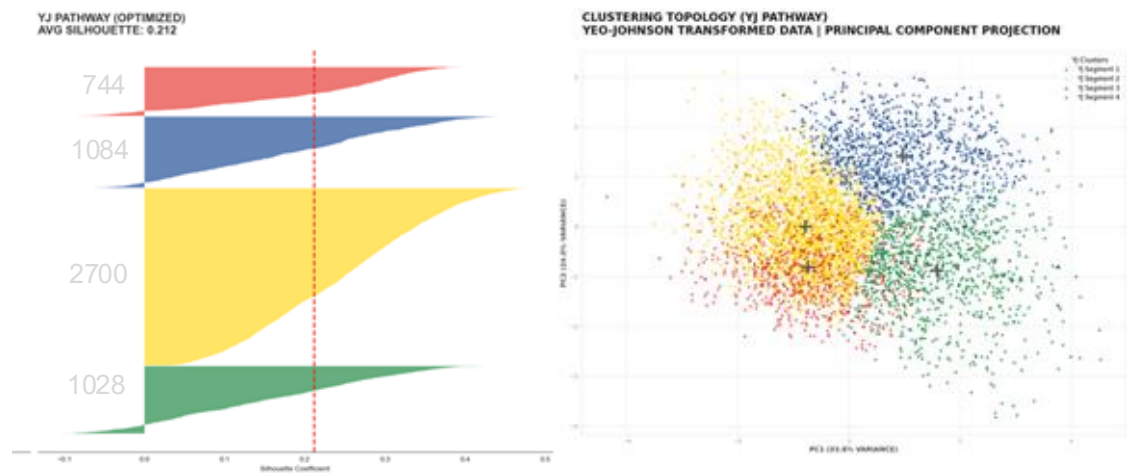
PATHWAY B (LG): RAW DATA TOPOLOGY EVOLUTION



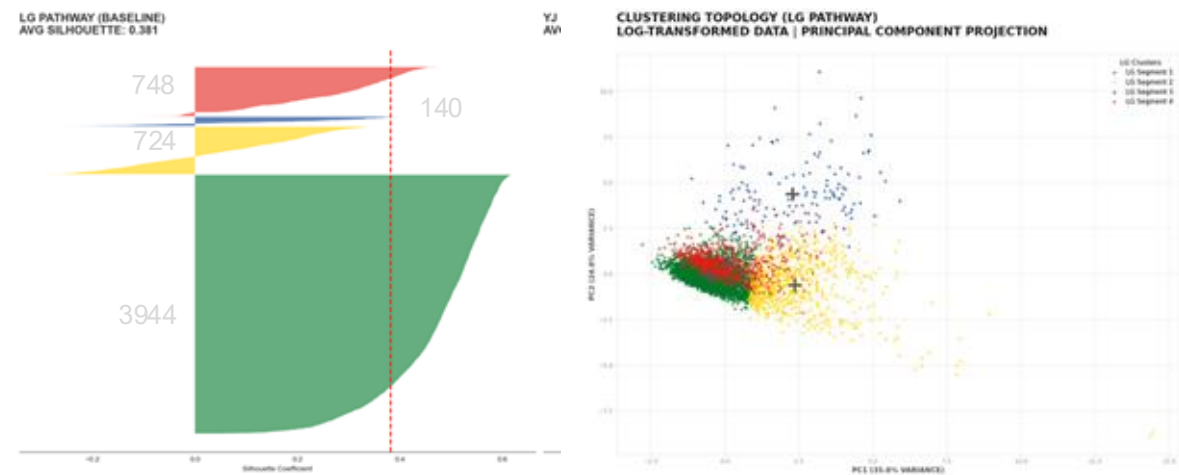
Log transform preserves magnitude differences — clusters form discrete islands with clearer separation than Pathway A. K=4 already isolates distinct behavioral groups; K=5 subdivides without adding new structure.

Evaluation and Performance

A: Yeo-Johnson Transform + Standardization | K=4



B: Log Transform + Standardization | K=4 🏆



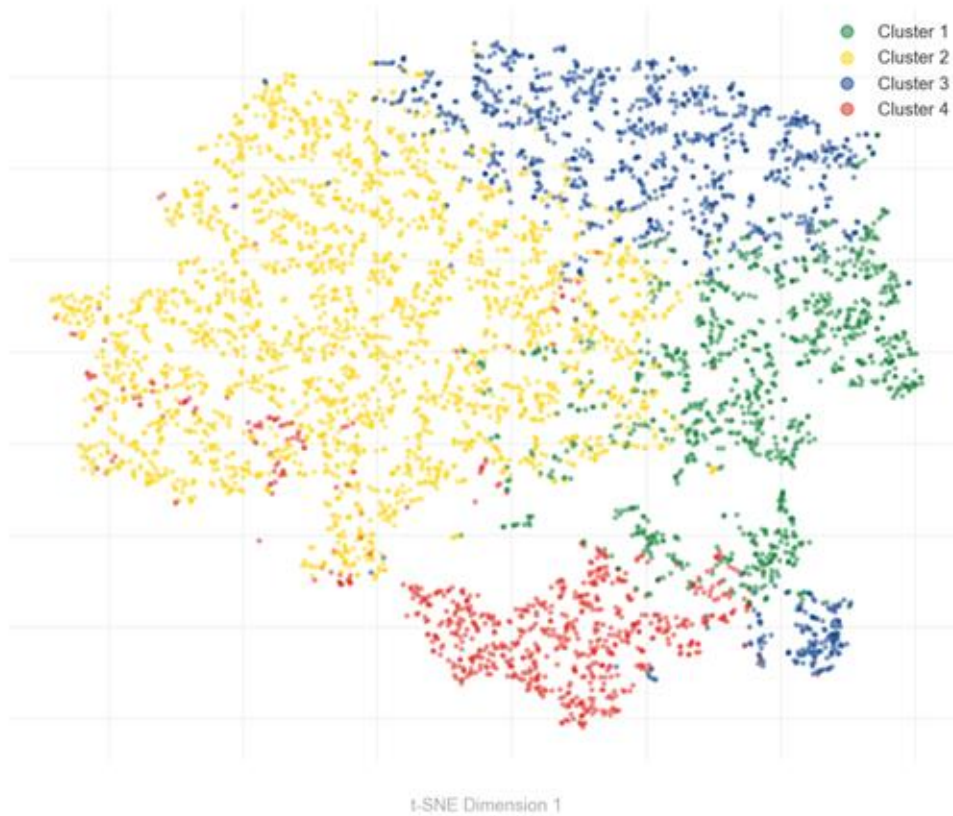
Metric	B [Log]	A[YJ]	Direction	Outcome
Silhouette Score	0.3814	0.2115	Higher is better	🏆 Pathway B (Tighter Clusters)
CH Index	1551.10	1285.74	Higher is better	🏆 Pathway B (Better Separation)
DB Index	1.1432	1.4800	Lower is better	🏆 Pathway B (Lower Overlap)

Model Selection: Log-pathway outperforms YJ on all three metrics (Silhouette +80%), producing tighter, better-separated clusters.

Why: Log preserves the magnitude of energy consumption; YJ over-normalizes and blurs boundaries between household types.

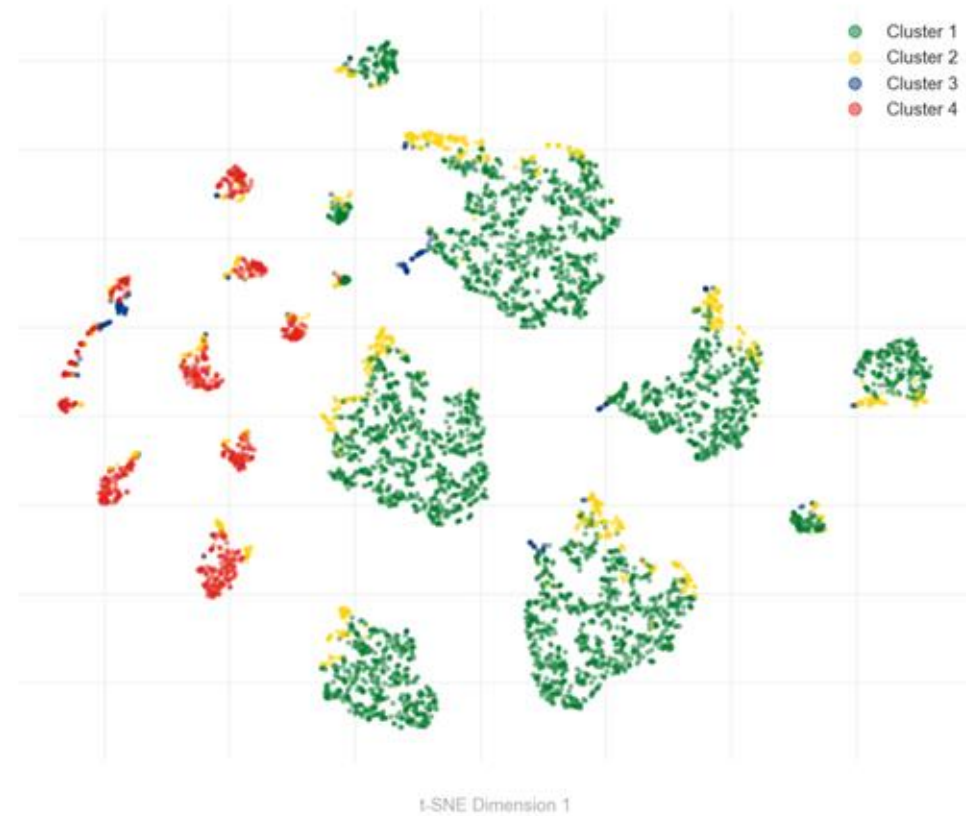
Cross-validation: ARI of 0.40 shows moderate structural overlap — the 4 personas are not artifacts of one method, but nor are the two paths interchangeable.

High-Dimensional Distribution Comparison



A: YJ Scaling

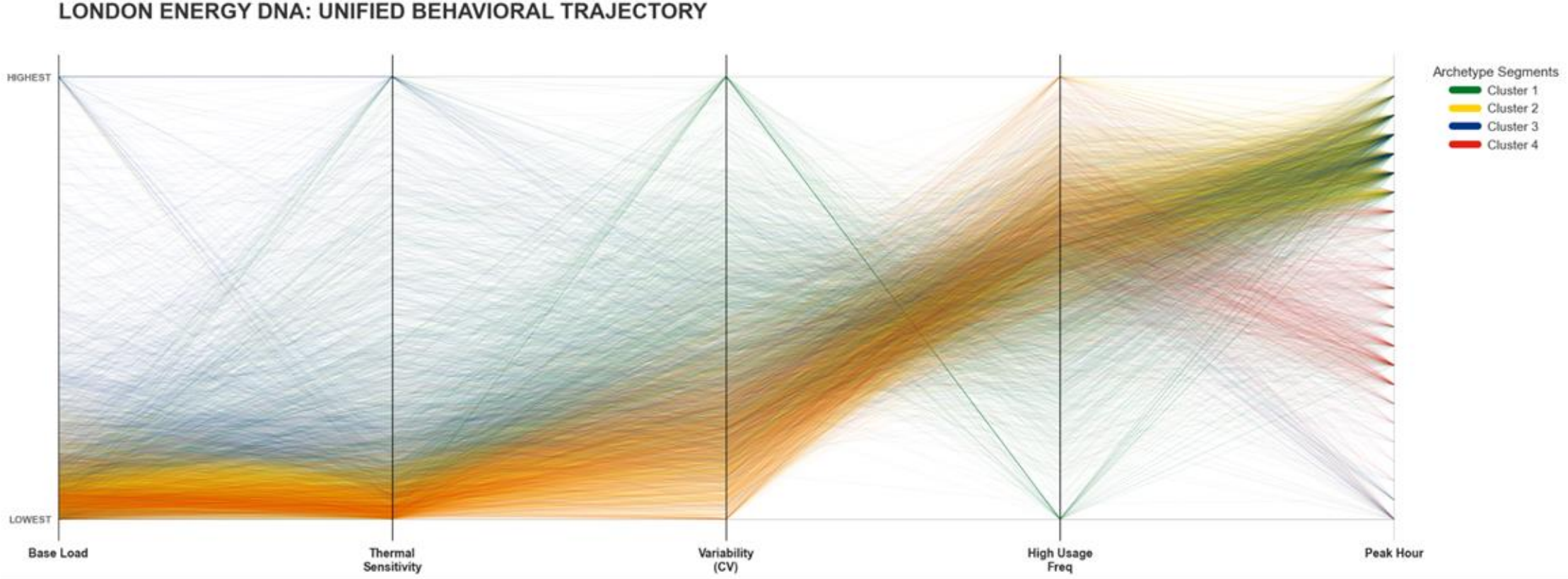
It exhibits a **Continuous Manifold**. Energy behavior exists on a spectrum, suggesting users transition smoothly between states.



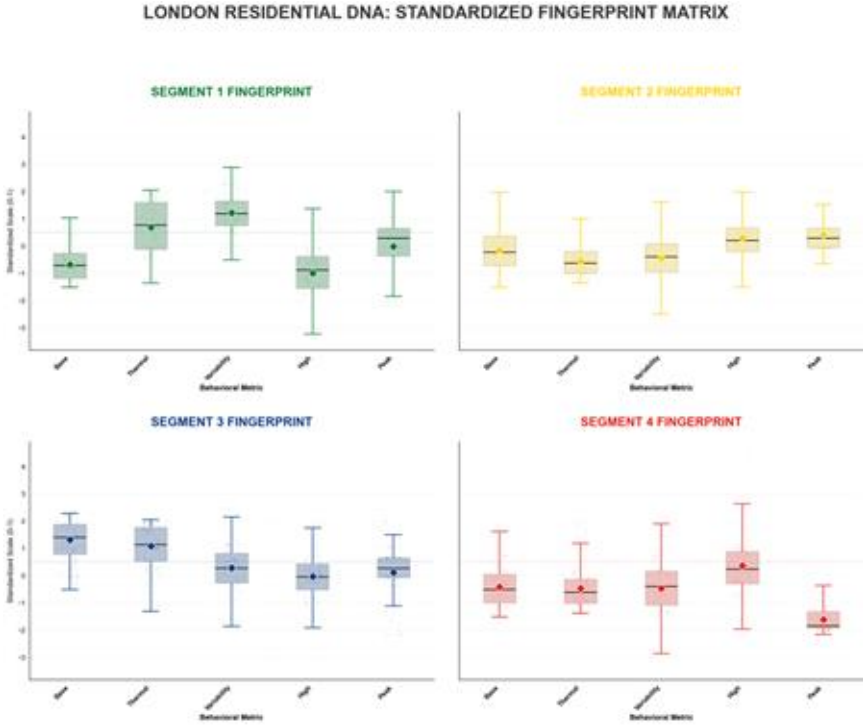
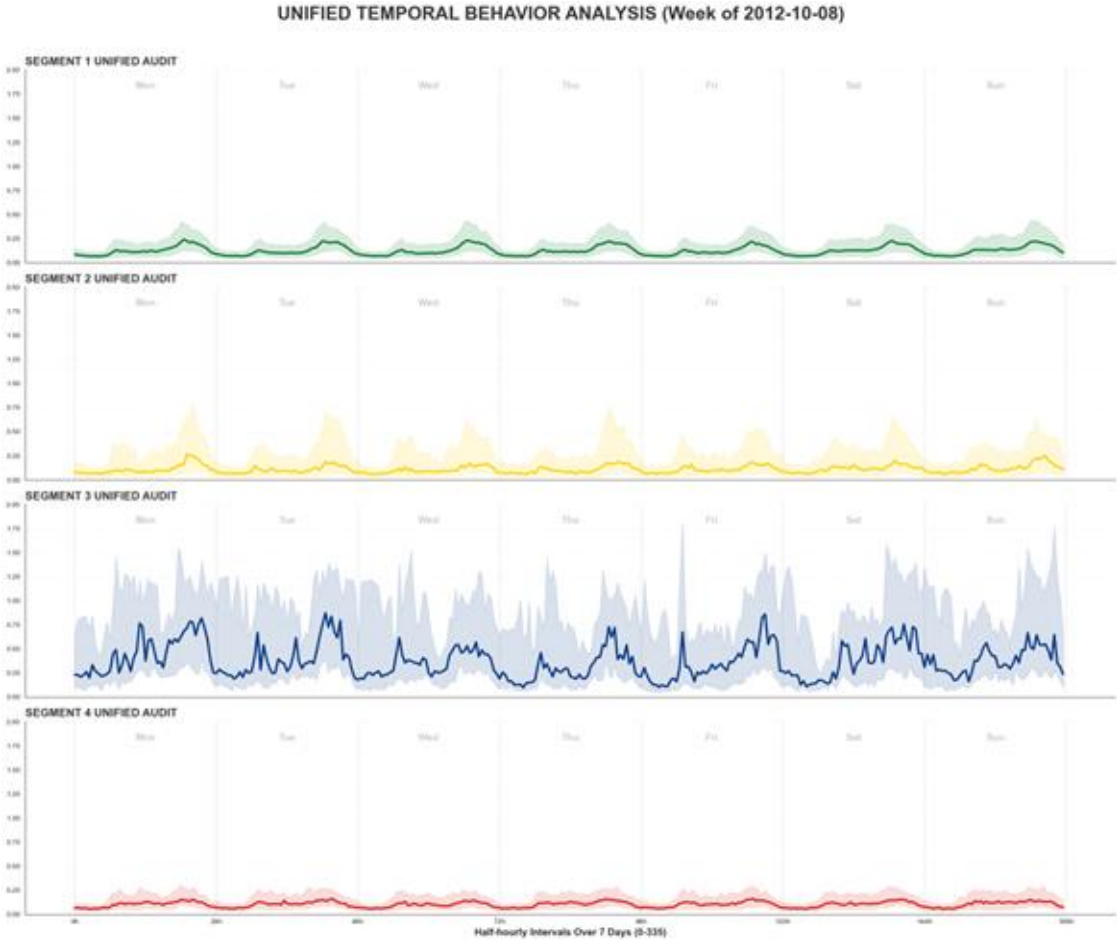
B: Log-Transform

It reveals **Discrete Micro-Communities**. The log-transform uncovers distinct "islands" of behavior, isolating specific user habits.

Parallel Coordinates: 4 Personas across 5 Behavioral Dimensions



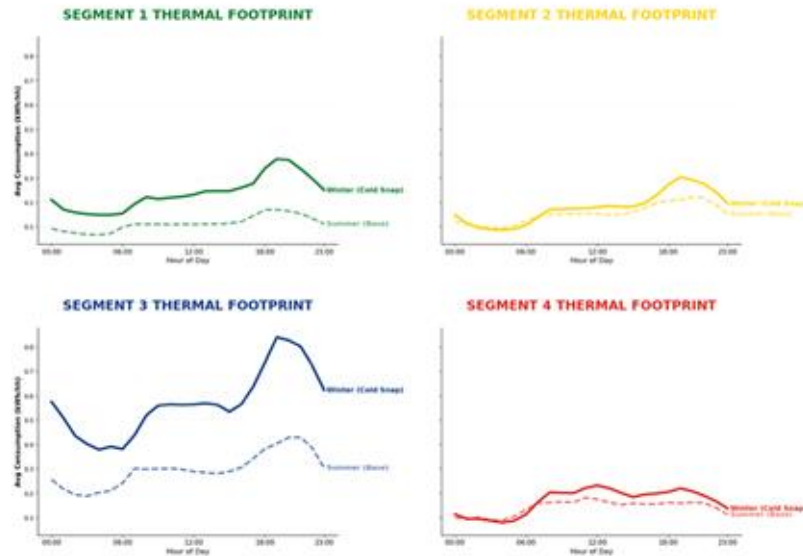
4 Personas: Temporal Profiles & Behavioral Fingerprints



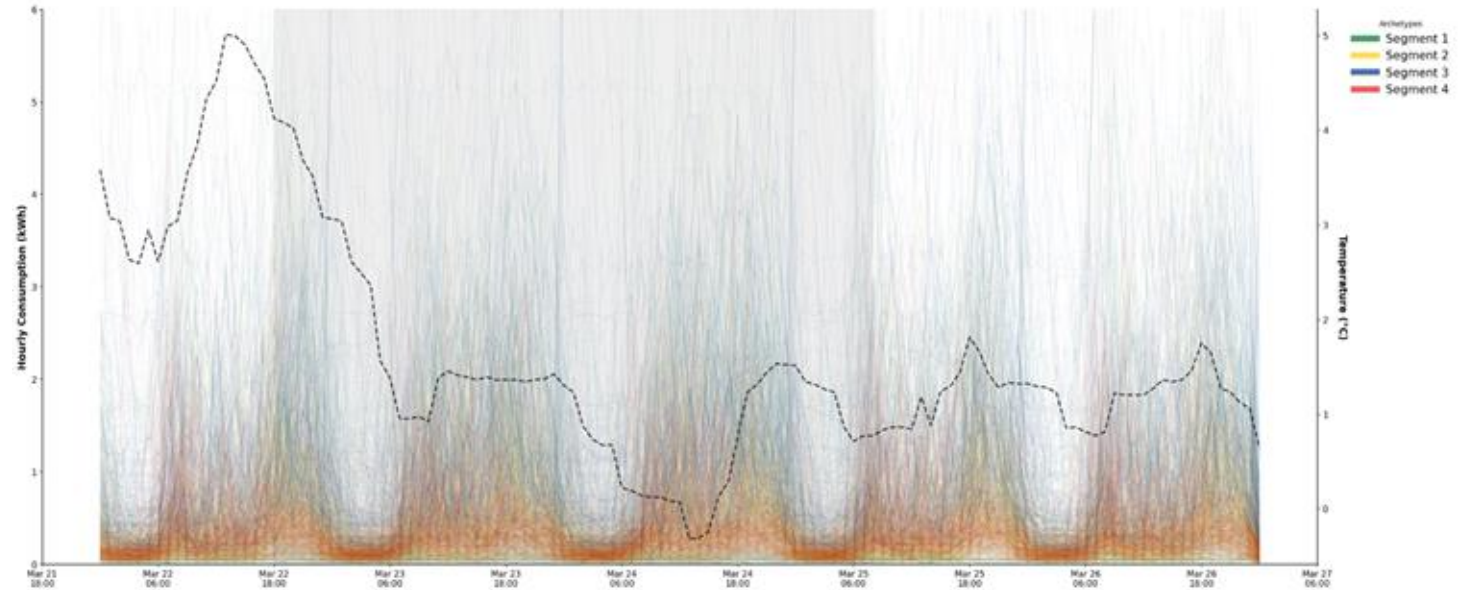
- S1 — Rhythmic Routines:** clear weekday/weekend contrast, moderate base load
- S2 — Standard Commuters:** flat profile, low thermal sensitivity
- S3 — High-occupancy Households:** highest consumption across all metrics
- S4 — Minimalist / Vacancy:** near-zero usage, likely unoccupied or under-metered

Thermal Sensitivity: Segment Profiles & Real-World Stress Test

THE LONDON SEASONAL AUDIT: BEHAVIORAL FINGERPRINTS
Direct Comparison of Winter Stress vs. Summer Baselines

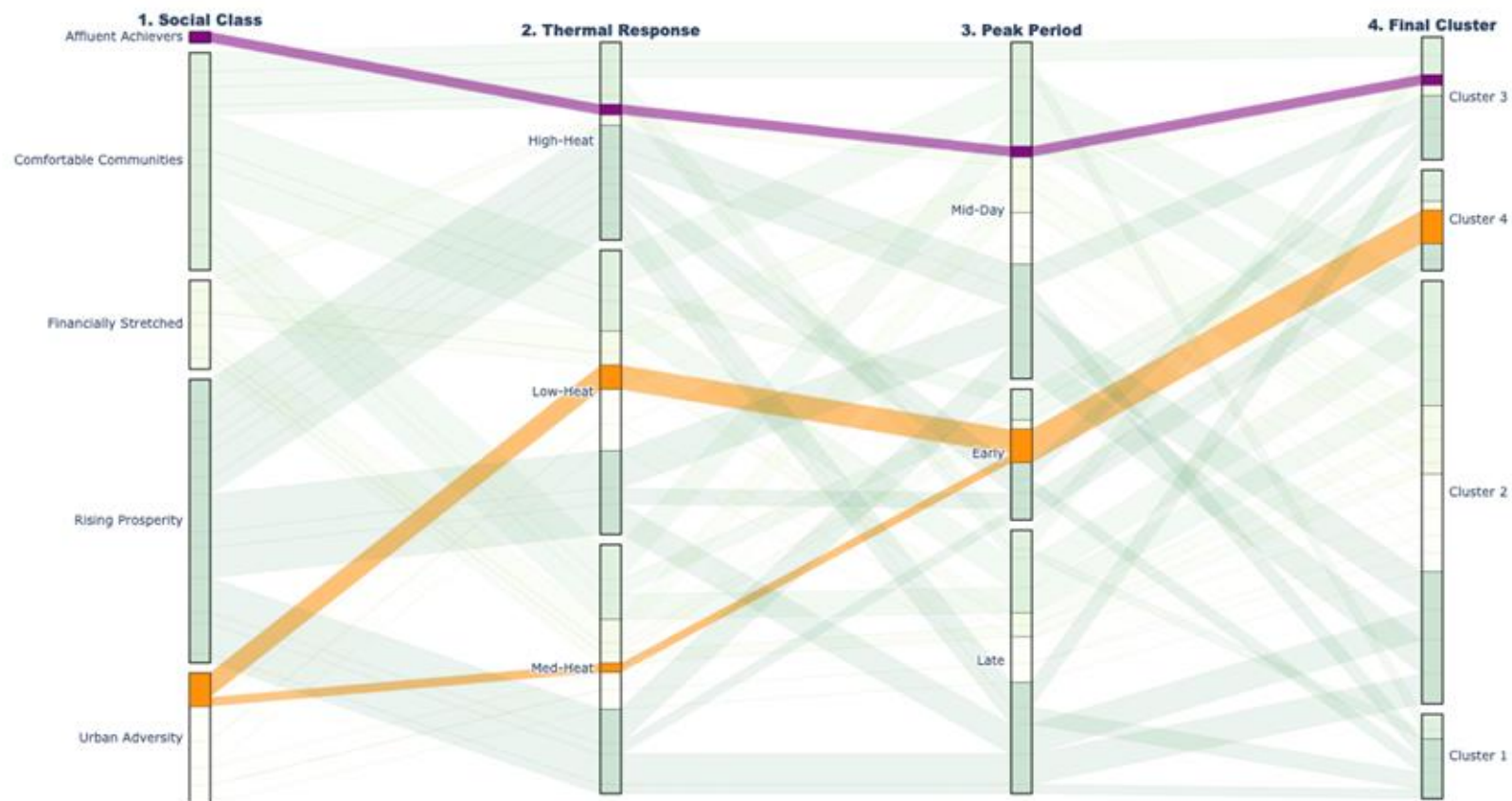


THE MARCH 2013 COLD SNAP: MICRO-BEHAVIORAL CLOUD ANALYSIS
1,000 Sampled Households Across London



S3 shows the widest winter-summer gap (left) and the sharpest surge when temperature drops below 0°C (right) — the most thermally sensitive persona.

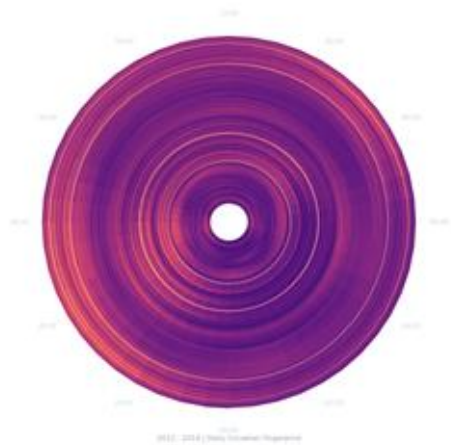
Energy Behavior Architecture



Social class does not map one-to-one onto energy behavior — affluent households and financially stretched households can converge into the same cluster through different thermal and temporal routes. The two highlighted pathways (purple, orange) represent the most policy-relevant contrasts.

Beyond Demographics: Household energy fingerprints, derived from smart meter data and refined through clustering, offer a closer proxy to actual living patterns than income or household size alone. This behavioral segmentation could inform more precise demand-side management, targeted retrofit programs, and equitable tariff design — intervening based on how people actually use energy, not who we assume they are.

Conclusion



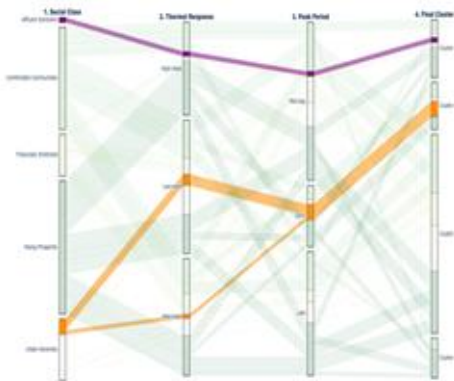
Energy Fingerprints as Behavioral Proxies

Moving from raw consumption data to a visual representation of human rhythm.



Embracing the Long-Tail Information

Retaining "extreme" signals to achieve higher contrast and distinct clustering results.



A Modular Framework for Granular Governance

Combining expert features with clustering to build a flexible observation network for lifestyle patterns.